Lesson notes · for you

Poisoned data

AI security · Upper secondary · Lesson 3

Time
20 min

Materials
Paper and a pen; Ten to twenty simple things to measure, such as leaves or beans, and a ruler; A phone or computer with Darsi, if you have one, for the lab and the code

Learning goals

- Explain data poisoning and backdoors.
- Find suspicious labels with a neighbour check in code.
- Decide what to remove, and record why.

Activity: Find the poison

1. Warm-up (3 min): Write down your guess: if two of twenty labels are wrong, will a model notice? Keep your guess to check later.
2. Measure (7 min): Measure ten things of two kinds (leaf or bean lengths) and label each one. Without looking, let a friend flip two of the labels.
3. Check (5 min): Sort the measurements from smallest to largest. Circle every label whose left and right neighbours both have the other label.
4. Share (5 min): Ask your friend which labels were flipped. Did you circle them? Did you circle an honest one at the edge of a group? Write down what you'd remove and why.

Measure harmless things, like leaves or beans. No device needed: the neighbour check works on paper.

Discussion

- Why can't a high test score prove the data is clean?
- Why should a person look before anything is removed?
- Where could someone plant a wrong label in your school's data?

Check yourself

- Explain the neighbour check in one sentence, and one thing it can get wrong.

Try it this week
With three friends, label the same ten pictures on your own, then compare. Where you disagree, talk it through and keep the majority, and notice which labels were careless.

Worksheet

Poisoned data

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Name _____ Class _____ Date _____

1 **New data was added last night. This morning the score on your locked test set fell from 96% to 80%. What do you do first?**

◯ Roll back, and check the new data ◯ Train again with even more data

◯ Delete the test set: it must be wrong

2 **A model can score well on all its normal tests and still have poisoned training data.**

◯ True ◯ False

3 **Three friends label the same photo. What does this print?**

```
votes = ["sick", "sick", "ok"]
print(votes.count("sick") > len(votes) / 2)
```

◯ False ◯ 2 ◯ True

4 **This version of suspects flags far too many leaves, including honest ones at the edge of each group. Which line has the mistake?**

Circle the line with the mistake. Write it correctly:

```
1  def suspects(xs, lab):
2      out = []
3      for i in range(1, len(lab) - 1):
4          left, me, right = lab[i-1:i+2]
5          if left != me or right != me:
6              out.append(xs[i])
7      return out
```

5 **Which of these help keep poison out of your data? Choose all that apply.**

☐ Add every free dataset you can find ☐ Three people label each example; keep the majority

☐ Delete every flagged example without looking ☐ Know where each dataset came from

☐ A locked, clean test set to compare scores

✎ **Think of data you or your class have collected: a survey, measurements, photos. How could a wrong or planted entry get in, and how would you spot it?**

Check your answers

Poisoned data

1. **Roll back, and check the new data**

   A sudden drop on a test set nobody else can touch points at the new data.

2. **True**

   A backdoor is built to pass normal tests: the model misbehaves only when the hidden trigger appears.

3. **True**

   Two of the three votes say "sick", and 2 is more than 3 / 2 = 1.5, so the majority wins and it prints True.

4. **Line 5:** `if left != me and right != me:`

   With or, one disagreeing neighbour is enough, so every leaf at the edge of a group gets flagged.

5. **Know where each dataset came from; Three people label each example; keep the majority; A locked, clean test set to compare scores**

   Knowing the source, majority labels and a locked test set all stop or reveal poison.